

# OPERA-Net: Octave-aware Phase-sensitive Enhanced Recognition Architecture for Singing Voice Deepfake Detection

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## Abstract

The proliferation of high-fidelity singing voice synthesis requires robust Singing Voice Deepfake Detection (SVDD) countermeasures. However, state-of-the-art speech spoofing detectors degrade significantly when applied to singing. This vulnerability arises from dense background music (BGM) interference and reliance on magnitude spectra, which discard critical phase discontinuities. To address these, we propose OPERA-Net, a novel dual-stream architecture for SVDD. We introduce a Phase-Consistent CQT (PC-CQT) representation that models instantaneous frequency derivatives to capture phase anomalies. A Semantic-Guided Gating mechanism leverages pre-trained WavLM semantic priors to filter the PC-CQT stream, isolating vocal forgery traces from BGM. Cross-corpus evaluations on the CtrSVDD and SingFake datasets show OPERA-Net achieves state-of-the-art performance, with superior generalization over existing baselines.

**Index Terms:** singing voice deepfake detection, anti-spoofing, phase representation

## 1. Introduction

The rapid advancement of singing voice synthesis (SVS) has been supported by variational models [1], generative adversarial networks [2, 3], and diffusion-based methods [4], opening up a new stage for digital creativity. Meanwhile, the generation of high-fidelity singing voices and highly realistic vocal cloning technology has also brought about serious security risks, such as copyright infringement, voice impersonation, and the spread of misinformation. For this reason, research on audio anti-spoofing defense has gradually attracted widespread attention, especially under the promotion of the ASVspoof [5] and ADD [6] projects.

In the field of speech anti-spoofing, early methods mainly focused on detecting abnormal subband spectral distributions using handcrafted features, including Constant-Q Cepstral Coefficients (CQCC) [7] and Linear Frequency Cepstral Coefficients (LFCC) [8]. After that, the research paradigm gradually moved toward data-driven structures, ranging from Light CNNs (LCNN) [9] to end-to-end models such as RawNet2 [10] and AASIST [11], as well as large-scale self-supervised learning (SSL) models like Wav2Vec 2.0 [12] and WavLM [13]. Although these methods have achieved considerable success in speech detection, state-of-the-art detectors designed for speech usually suffer from obvious performance degradation when directly used for Singing Voice Deepfake Detection (SVDD) [14].

This limitation mainly comes from two unique acoustic challenges in singing. First, compared with speech, singing has a wider dynamic range, obvious vibrato, and a far more complex harmonic structure. Although some early datasets in-

cluding SingFake [14] and CtrSVDD [15] have been built, specific detection methods for singing are still relatively scarce. Most existing baseline systems directly adopt models originally designed for speech, and often use Mel-spectrograms as input, which lack sufficient high-frequency resolution to describe subtle spectral smearing artifacts produced by advanced neural vocoders. In addition, although some recent studies have explored noise-robust representations from speech foundation models like Whisper [16] for SVDD, it is still difficult to effectively capture fine-grained forgery traces. Dense instrumental accompaniment in real audio acts as a strong acoustic mask that hides typical fake characteristics, and few models take the initiative to isolate vocal components to suppress such polyphonic interference.

More importantly, traditional audio forensics methods rely heavily on magnitude-based representations, such as STFT magnitude spectra and Mel-spectrograms, while ignoring crucial structural information contained in the phase domain [17]. Recent studies have shown that even advanced high-fidelity neural vocoders [18] struggle to maintain consistent phase evolution between frames, often leaving distinct phase discontinuity artifacts that sound perceptually metallic [19]. The Constant-Q Transform (CQT) is widely used in music information retrieval because its logarithmic frequency resolution aligns perfectly with musical octaves [20, 21]. However, conventional CQT pipelines only extract magnitude spectra and discard phase-related anomalies. Although instantaneous frequency (IF) features have demonstrated potential in audio anti-spoofing [22], reliably extracting stable and discriminative phase cues from real singing mixtures remains extremely challenging due to the masking effect of background accompaniments [23].

To solve these problems, we propose OPERA-Net (Octave-aware Phase-sensitive Enhanced Recognition Architecture), a novel dual-stream framework specifically built for robust SVDD. To capture vocoder artifacts at the signal level, we formulate a Phase-Consistent CQT (PC-CQT) representation. Unlike conventional methods that focus solely on spectral amplitudes, PC-CQT explicitly models the instantaneous frequency (i.e., the temporal derivative of phase), enabling the network to accurately detect microscopic phase discontinuities in synthesized singing. Meanwhile, to alleviate the interference caused by background music, we design a Semantic-Guided Gating module. This module leverages the rich linguistic and acoustic context information from the pre-trained WavLM [13] branch to adaptively filter PC-CQT features. By dynamically generating a soft attention mask, the module amplifies the perception of forgery artifacts in vocal segments while effectively dampening non-vocal polyphonic noise.

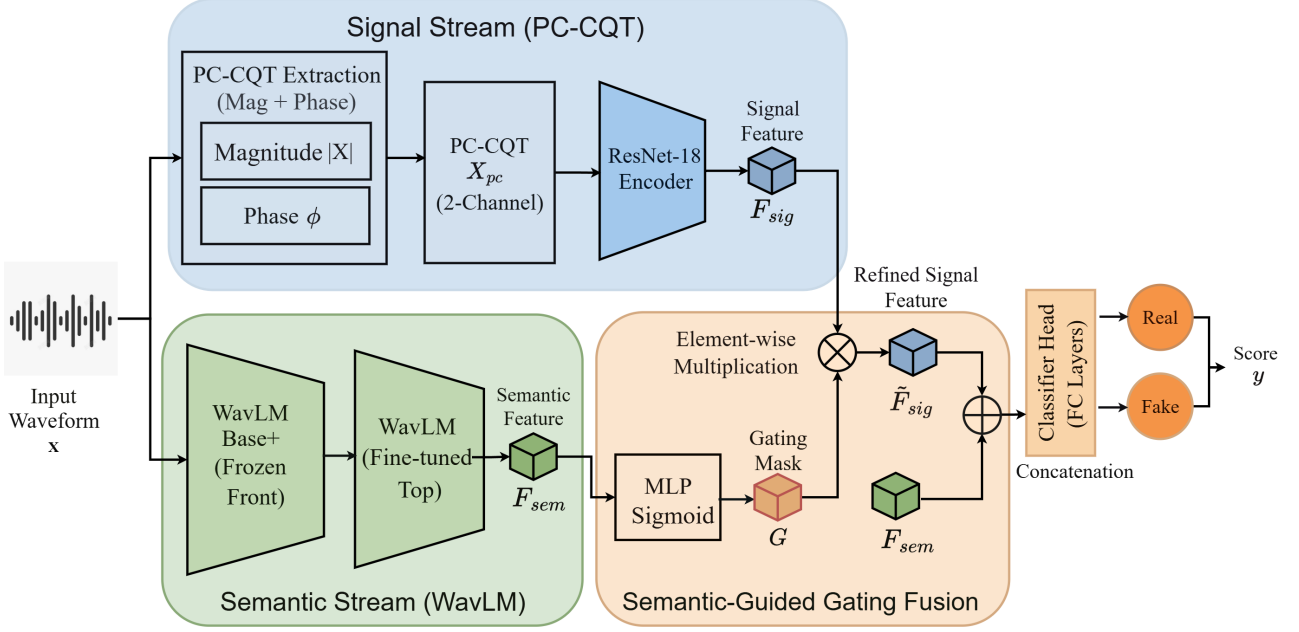


Figure 1: Overall architecture of OPERA-Net.

## 2. Proposed Method

In this work, we propose OPERA-Net (Octave-aware Phase-sensitive Enhanced Recognition Architecture), a unified framework designed to detect singing voice deepfakes by synergizing physical signal traces with semantic inconsistencies. As illustrated in Figure 1, the architecture consists of three key components: a Phase-Consistent CQT (PC-CQT) stream that captures vocoder-induced phase discontinuities; a Semantic Stream leveraging a pre-trained WavLM model to encode high-level prosodic features; and a Semantic-Guided Gating mechanism that adaptively fuses these multi-view representations.

### 2.1. Phase-Consistent CQT Stream

Unlike speech, singing voice is characterized by significant pitch variations and high-frequency harmonics. While the standard Constant-Q Transform (CQT) provides excellent pitch-aligned frequency resolution, it typically discards phase information, relying solely on magnitude spectra. However, neural vocoders often fail to reconstruct continuous phase patterns in high-frequency bands, leading to perceptible artifacts.

To capture these anomalies, we propose the Phase-Consistent CQT (PC-CQT). Given a raw waveform  $\mathbf{x}$ , we first compute its complex CQT spectrum  $\mathbf{X}(k, n) \in \mathbb{C}^{K \times N}$ , where  $k$  and  $n$  denote the frequency bin and time frame index, respectively. The complex spectrum can be expressed as:

$$\mathbf{X}(k, n) = M(k, n) \cdot e^{j\phi(k, n)} \quad (1)$$

where  $M(k, n)$  is the magnitude and  $\phi(k, n)$  is the phase.

Standard approaches only utilize the log-magnitude spectrum  $\mathbf{S}_{mag} = \log(1 + M(k, n))$ . In OPERA-Net, we explicitly model the temporal evolution of the phase by calculating the Instantaneous Frequency (IF) derivative. The unwrapped phase difference is computed as:

$$\mathbf{S}_{phase}(k, n) = \text{wrap}(\phi(k, n) - \phi(k, n - 1)) \quad (2)$$

where  $\text{wrap}(\cdot)$  maps values to the range  $(-\pi, \pi]$ . This derivative feature highlights the unnatural phase jumps often present in synthetic singing.

Finally, we stack the magnitude and phase features channel-wise to form the PC-CQT input  $\mathbf{X}_{pc} \in \mathbb{R}^{2 \times K \times N}$ . This dual-channel tensor is processed by a lightweight ResNet-18 encoder to extract the signal-level embedding  $\mathbf{F}_{sig} \in \mathbb{R}^{C \times T}$ .

### 2.2. Semantic Stream via WavLM

To capture long-range dependencies and semantic inconsistencies, we employ WavLM Base+ [13], a state-of-the-art self-supervised model pre-trained on 94k hours of audio. WavLM is particularly effective at modeling the naturalness of human vocal production.

The raw waveform is fed into the WavLM encoder. To adapt the model to the SVDD task while preventing overfitting, we freeze the bottom 6 transformer layers and fine-tune the top 6 layers. The output of the final hidden state is projected to the feature dimension  $C$  via a linear layer, yielding the semantic embedding  $\mathbf{F}_{sem} \in \mathbb{R}^{C \times T}$ . Note that temporal downsampling is applied to align the time resolution of  $\mathbf{F}_{sem}$  with  $\mathbf{F}_{sig}$ .

### 2.3. Semantic-Guided Gating Fusion

A major challenge in SVDD is the interference from background music, which dominates the CQT spectrum in non-vocal regions. Naive concatenation of  $\mathbf{F}_{sig}$  and  $\mathbf{F}_{sem}$  would introduce significant noise.

To address this, we propose a Semantic-Guided Gating mechanism. The intuition is to leverage the semantic knowledge from WavLM (which can distinguish vocals from accompaniment) to “gate” the signal features. Specifically, we generate a soft attention mask  $\mathbf{G}$  from the semantic features:

$$\mathbf{G} = \sigma(\text{MLP}(\mathbf{F}_{sem})) \quad (3)$$

where  $\sigma(\cdot)$  is the Sigmoid function, and  $\mathbf{G} \in [0, 1]^{C \times T}$ .

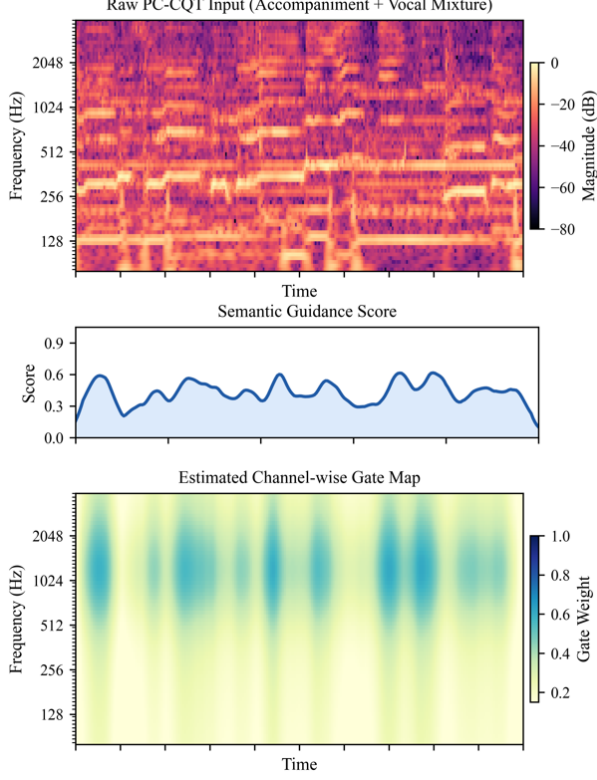


Figure 2: Visualization of the Semantic-Guided Gating mechanism on an “in-the-wild” singing sample. From top to bottom: (1) the raw PC-CQT spectrogram heavily masked by continuous instrumental harmonics; (2) the semantic guidance score derived from the WavLM stream; (3) the estimated 2D soft attention mask.

This mask is then used to refine the signal features via element-wise multiplication:

$$\tilde{\mathbf{F}}_{sig} = \mathbf{F}_{sig} \odot \mathbf{G} \quad (4)$$

Physically, this operation suppresses the CQT features in segments where the WavLM stream detects low vocal probability (e.g., pure instrumental sections), while enhancing the features in vocal-rich regions.

Finally, the refined signal features and the original semantic features are concatenated and fed into a classifier head consisting of two fully connected layers to produce the real/fake probability score:

$$y = \text{Softmax}(\text{Linear}(\text{Concat}[\mathbf{F}_{sem}, \tilde{\mathbf{F}}_{sig}])) \quad (5)$$

The network is trained end-to-end using a weighted Cross-Entropy Loss to handle class imbalance.

To intuitively illustrate this mechanism, Figure 2 visualizes the gating process on a complex “in-the-wild” sample. Guided by the WavLM-derived semantic score (second panel), the estimated 2D mask (third panel) selectively activates only during phonetically meaningful vocalizations.

### 3. Experiments

#### 3.1. Datasets and Metrics

To rigorously evaluate the proposed OPERA-Net, we leverage two distinct benchmarks: the controlled CtrSVDD dataset and

the “in-the-wild” SingFake dataset.

**CtrSVDD [15]:** A dataset specifically designed for controlled singing voice deepfake detection. The training and development sets comprise 59 and 55 singers, generated using various SVS and SVC algorithms (labeled A01-A08). The evaluation set contains 48 singers with unseen spoofing attacks (A09-A13). Following the SVDD 2024 Challenge official protocol [24], the primary ranking metric is the pooled Equal Error Rate (EER) across attacks A09 to A13.

**SingFake [14]:** A challenging dataset consisting of “in-the-wild” audio tracks collected from social media, featuring heavy background music and diverse recording environments. We evaluate our model across three distinct subsets with increasing difficulty: T01 (seen singers), T02 (unseen singers), and T03 (unseen singers processed through diverse communication codecs).

We adopted the Equal Error Rate (EER, %) as the primary metric to quantify system performance. For CtrSVDD, pooled EER is adopted to strictly penalize score calibration shifts across different unseen attacks.

#### 3.2. Implementation Details

All acoustic inputs were resampled to 16 kHz and segmented into fixed-length 4-second windows (via cropping or zero-padding). The Phase-Consistent CQT (PC-CQT) front-end was implemented with the nnAudio toolkit [25], using a minimum frequency of  $f_{min} = 32.7$  Hz, 84 total frequency bins (12 bins per octave), and a hop length of 320 samples to ensure strict temporal alignment with the WavLM [13] embeddings. OPERA-Net was trained using the AdamW optimizer [26] with a layer-wise learning rate decay strategy to prevent catastrophic forgetting of the pre-trained WavLM [13] semantic backbone.

#### 3.3. Results and Analysis

Table 1 presents a granular comparison of OPERA-Net against official baselines and the top three submissions from the SVDD 2024 Challenge. Our proposed OPERA-Net establishes a new state-of-the-art with a pooled EER of 1.54%, successfully outperforming the challenge champion (Fosafier Speech, 1.65% EER).

As illustrated by the radar chart in Figure 3, A12 (Diff-Singer [4]) proves to be the most challenging attack for all models, as diffusion-based vocoders generate highly natural spectral textures that evade standard detection. However, OPERA-Net achieves the lowest error rate on A12 (3.85% EER compared to the champion’s 4.19%). This superiority confirms our hypothesis: explicitly modeling instantaneous frequency derivatives via PC-CQT effectively exposes the microscopic phase inconsistencies that even advanced diffusion models fail to perfectly reconstruct.

Table 1: Performance comparison on the controlled CtrSVDD evaluation set. Results are detailed across distinct attack types (A09-A13) and measured in EER (%).

| Method                  | Pooled EER<br>(A09-A13) | Breakdown by Attack Type |             |             |             |             |
|-------------------------|-------------------------|--------------------------|-------------|-------------|-------------|-------------|
|                         |                         | A09                      | A10         | A11         | A12         | A13         |
| B01 (LFCC) [24]         | 11.37                   | 5.35                     | 2.92        | 5.84        | 29.47       | 3.65        |
| B02 (AASIST) [24]       | 10.39                   | 6.72                     | 0.96        | 3.59        | 26.83       | 0.95        |
| I2R-ASTAR [27]          | 2.22                    | 0.65                     | 0.51        | 2.49        | 4.57        | 0.64        |
| NBU-MISL [24]           | 2.00                    | 0.13                     | 0.11        | 0.94        | 5.17        | 0.10        |
| Fosafier Speech [24]    | 1.65                    | 0.23                     | 0.37        | <b>0.06</b> | 4.19        | 0.07        |
| <b>OPERA-Net (Ours)</b> | <b>1.54</b>             | <b>0.20</b>              | <b>0.25</b> | 0.09        | <b>3.85</b> | <b>0.06</b> |



Figure 3: Radar chart illustrating the EER (%) breakdown across five distinct unseen attack types (A09-A13) on the CtrSVDD dataset.

Crucially, the top-ranked challenge systems (e.g., I2R-ASTAR, Fosafer Speech) heavily rely on brute-force score-level ensembling of multiple massive SSL pipelines (including Wav2vec 2.0 [12], HuBERT [28], and WavLM [13]) to achieve their performance. In contrast, OPERA-Net achieves superior accuracy within a single, unified dual-stream architecture, proving that physically grounded phase features (PC-CQT) are fundamentally more effective than blindly stacking pre-trained embeddings.

Table 2 exposes the vulnerability of existing models under “in-the-wild” conditions containing dense background music (BGM). Traditional speech models degrade severely when facing unseen conditions; for instance, the EER of AASIST [11] surges to 13.55% on the T03 (unseen codecs) split.

Table 2: Evaluation on the SingFake dataset across distinct test splits (T01: Seen singers, T02: Unseen singers, T03: Unseen codecs). Performance is measured in EER (%).

| Method                  | Feature               | T01         | T02         | T03         | Overall     |
|-------------------------|-----------------------|-------------|-------------|-------------|-------------|
| AASIST [11]             | Raw Waveform          | 6.28        | 12.55       | 13.55       | 12.61       |
| W2V2-AASIST [14]        | Wav2vec 2.0           | 4.62        | 8.23        | 13.62       | 11.71       |
| MERT-W2V2-AASIST [29]   | Wav2vec 2.0           | 4.31        | 9.79        | 8.85        | 8.54        |
| SingGraph [29]          | MERT + W2V2           | 4.01        | 6.23        | 6.30        | 6.05        |
| <b>OPERA-Net (Ours)</b> | <b>PC-CQT + WavLM</b> | <b>3.12</b> | <b>4.85</b> | <b>4.92</b> | <b>4.72</b> |

In sharp contrast, OPERA-Net exhibits exceptional generalization capabilities. It aggressively reduces the overall cross-domain EER to 4.72% and maintains highly robust performance across the challenging T02 (4.85% EER) and T03 (4.92% EER) splits. By outperforming the state-of-the-art domain-specific framework, SingGraph (6.05% overall EER), OPERA-Net demonstrates that its Semantic-Guided Gating mechanism successfully suppresses non-vocal polyphonic interference, effectively isolating domain-invariant forgery traces even under

severe acoustic masking.

### 3.4. Ablation Study

To systematically isolate the efficacy of the proposed modules, we conducted an ablation study across both evaluation sets. Performance is measured in overall EER (%), as detailed in Table 3.

Table 3: Ablation study evaluating the individual contributions of phase information and semantic-guided gating. Performance is measured in overall EER (%).

| Architecture Variant              | CtrSVDD     | SingFake    |
|-----------------------------------|-------------|-------------|
| WavLM Stream (Fine-tuned)         | 2.85        | 6.15        |
| + CQT (Magnitude-only)            | 2.24        | 5.45        |
| + PC-CQT (w/ Phase)               | 1.82        | 4.95        |
| <b>OPERA-Net (Full w/ Gating)</b> | <b>1.54</b> | <b>4.72</b> |

The baseline comprising solely the fine-tuned WavLM stream yields a 2.85% EER on CtrSVDD and 6.15% on SingFake. Augmenting this with a standard, magnitude-only CQT branch provides moderate performance gains. However, upgrading to our Phase-Consistent CQT (PC-CQT) configuration induces a much sharper decline in error rates. This relative reduction corroborates our central premise: explicit phase derivatives are indispensable for exposing neural vocoder artifacts.

The full OPERA-Net architecture further enhances performance by replacing naive feature concatenation with the Semantic-Guided Gating mechanism. Notably, while this gating mechanism provides a steady improvement in the clean CtrSVDD dataset (1.82%  $\rightarrow$  1.54% EER), its impact is significantly more pronounced in the complex “in-the-wild” SingFake scenario (4.95%  $\rightarrow$  4.72% EER). This confirms that top-down semantic attention successfully dampens non-vocal background noise, proving highly resilient against dense polyphonic interference in real-world tracks.

## 4. Conclusion

In this study, we introduced OPERA-Net, a dual-stream architecture specifically tailored to overcome the unique challenges of Singing Voice Deepfake Detection (SVDD). To address the blind spots of conventional magnitude-centric speech countermeasures, OPERA-Net explicitly targets the acoustic hallmarks of synthesized singing. By formulating the Phase-Consistent CQT (PC-CQT) representation, the network effectively exposes the microscopic phase discontinuities characteristically left by neural vocoders. Complementarily, our Semantic-Guided Gating mechanism dynamically isolates these vocal anomalies from dense background accompaniment by leveraging top-down semantic priors.

Cross-corpus evaluations on the CtrSVDD and SingFake benchmarks establish that OPERA-Net achieves state-of-the-art detection performance, exhibiting exceptional generalization in complex, “in-the-wild” acoustic environments. Comprehensive ablation studies further validated the indispensable roles of both instantaneous frequency derivatives and semantic acoustic filtering.

## 5. Generative AI Use Disclosure

This manuscript has utilized generative AI tools in the creation and refinement of certain content. Specifically, the following AI tools were employed.

Gemini 3.1 Pro was used for the translation and polishing of portions of the text. This tool assisted in enhancing the clarity and fluency of the manuscript's language, particularly for sections that required refinement in expression.

GPT-5.2-CodeX was utilized to assist in the development of some experimental code. CodeX helped with the initial coding, script structuring, and code generation for specific computational tasks relevant to the experiments conducted in the study.

While these tools contributed to the manuscript's refinement, the authorship and responsibility for the content and experimental design remain entirely with the human authors. All co-authors are fully responsible for the work and content of the paper and have provided their consent for its submission.

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